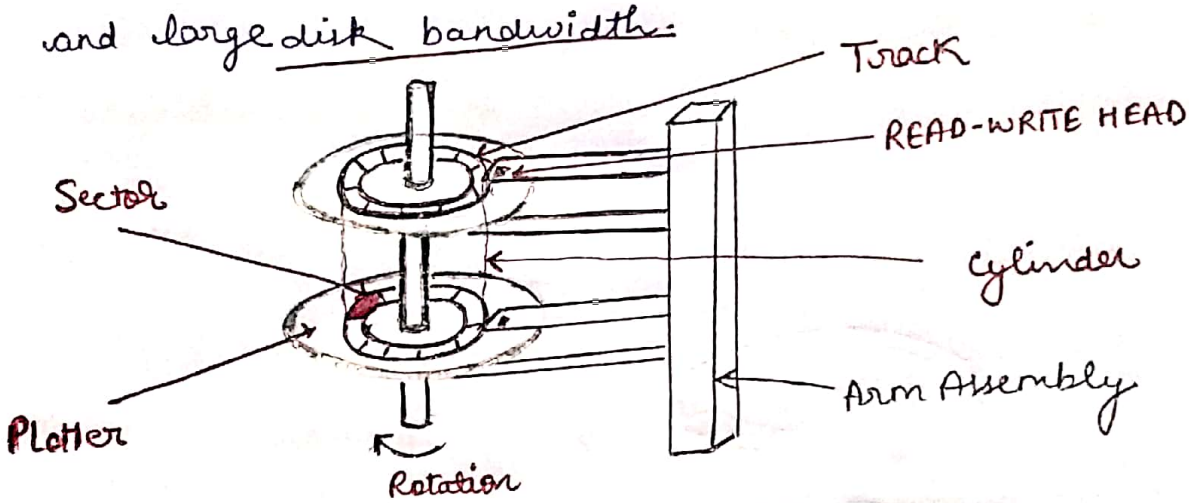
SECONDARY STORAGE STRUCTURE

Ques. What is disk scheduling?

Ans. one of the major responsibility of OS is to use hardware efficiently.

Thus, disk drives should have fast access time and large disk bandwidth.



Access time consist of
- → SEEK TIME
- → ROTATIONAL LATENCY

↳ Seek time is the time for disk arm to move read - write head to the cylinder containing desired sector.

↳ Rotational latency is the additional time for disk to rotate desired sector to disk head.

$$\text{Disk Bandwidth} = \frac{\text{total number of Bytes transferred}}{\text{total time between first request for service and the completion of last transfer.}}$$

* Access time and bandwidth can be improved by managing the order in which disk input and output requests are serviced.

⇩

When one request is completed, the operating system chooses which pending request to service next. For this choice several disk-scheduling algorithms are used -

1. FCFS Scheduling
2. SSTF Scheduling
3. SCAN scheduling
4. C-SCAN scheduling
5. LOOK Scheduling

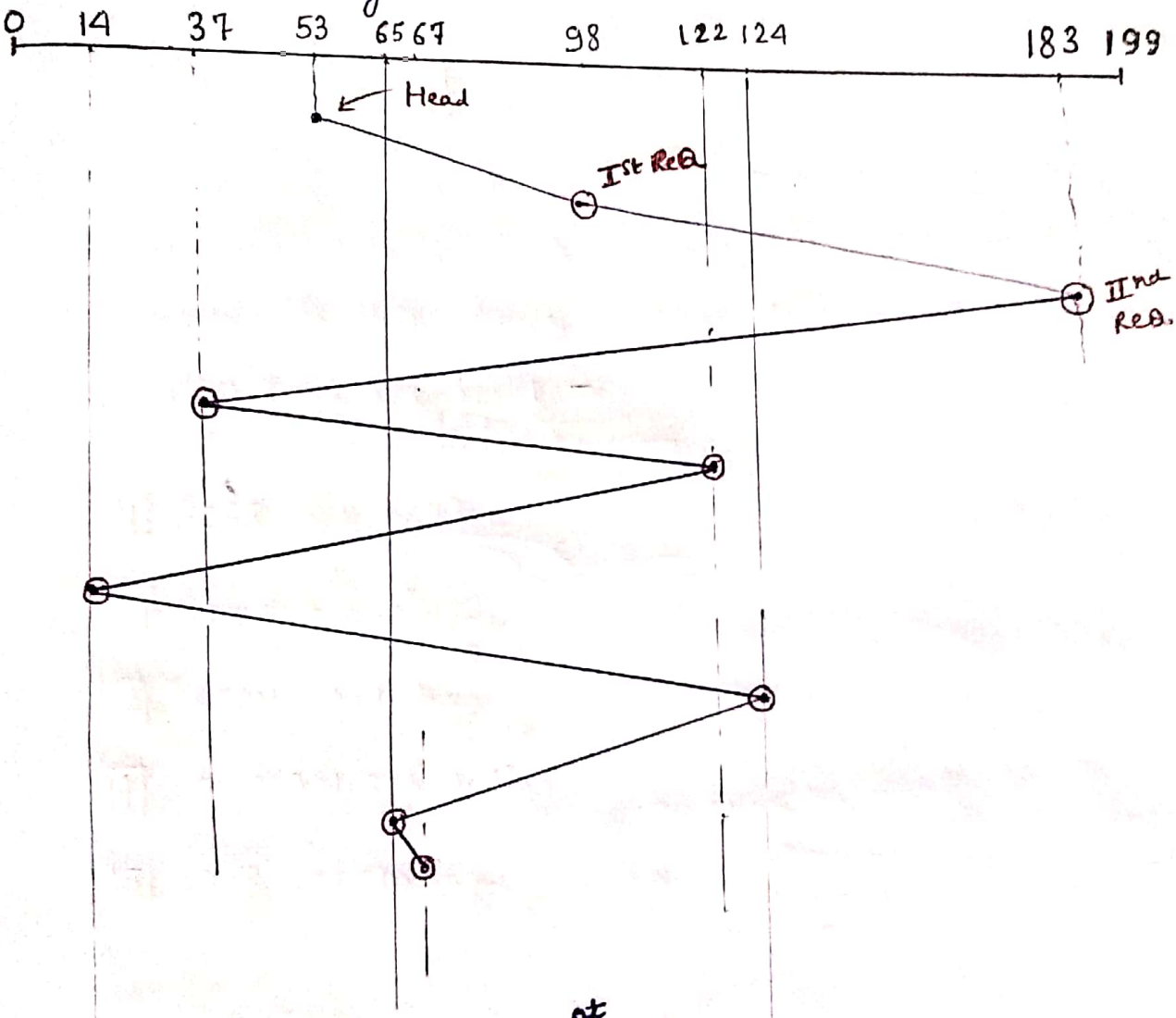
① **FCFS Scheduling** - Not the fastest service.

- The cylinder request coming first is serviced first
- Considering disk Queue with requests for I/O to blocks on cylinders

Queue - 98, 183, 37, 122, 14, 124, 65, 67

Head starts at - 53

For this illustration, we are considering there are 0 to 199 cylinders.



- Here Starting head is at 53rd and first Request is 98th cylinder

  ∴ Total Head movement = |53 - 98| = 45 cylinders

- Similarly next request is 183 and previous cylinder was 98th

  ∴ Total head movement = |98 - 183| = 85

- Similarly, |183 - 37| = 146

  |37 - 122| = 85

  |122 - 14| = 108

$|14-124| = 110$

$|124-65| = 59$

$|65-67| = 2$

$\therefore$ Total head movement $= 45 + 85 + 146 + 85 + 108 + 110 + 59 + 2$

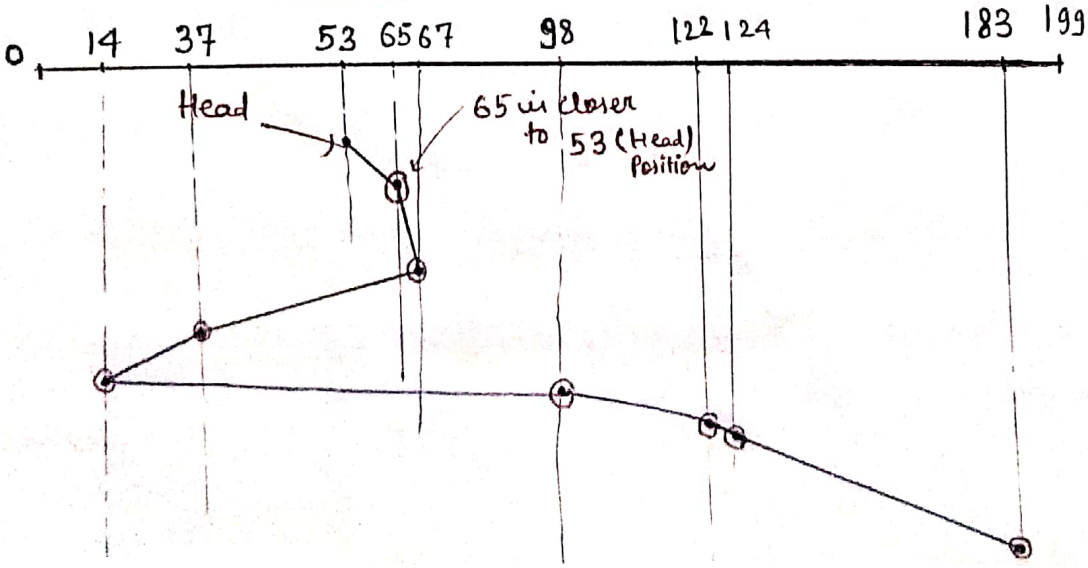
$= 640$ Cylinders

② SSTF Scheduling -

- In Shortest Seek time first scheduling, requests close to head position are serviced one-by-one.
- This may also cause starvation of some requests.
- Considering the same requests' example.

Queue - 98, 183, 37, 122, 14, 124, 65, 67

Head starts at - 53



$|53-65| = 12$

$|65-67| = 2$

$|67-37| = 30$

$|37-14| = 23$

$|14-98| = 84$

$|98-122| = 24$

$|122-124| = 2$

$|124-183| = 59$

$\therefore$ Total head movement $= 12+2+30+23+84+24+2+59$

$= 236$ Cylinders

③ <u>SCAN Scheduling</u>

- Head continuously scan back & forth across the disk.
- Also called Elevator

↓

As disk arm behaves just like an elevator i.e. Servicing requests all the way up and then reversing the service of requests.

This scheduling required to Know the head position & Head movement.

<u>Conclusions</u>

① Request arriving in front of head will be serviced immediately.

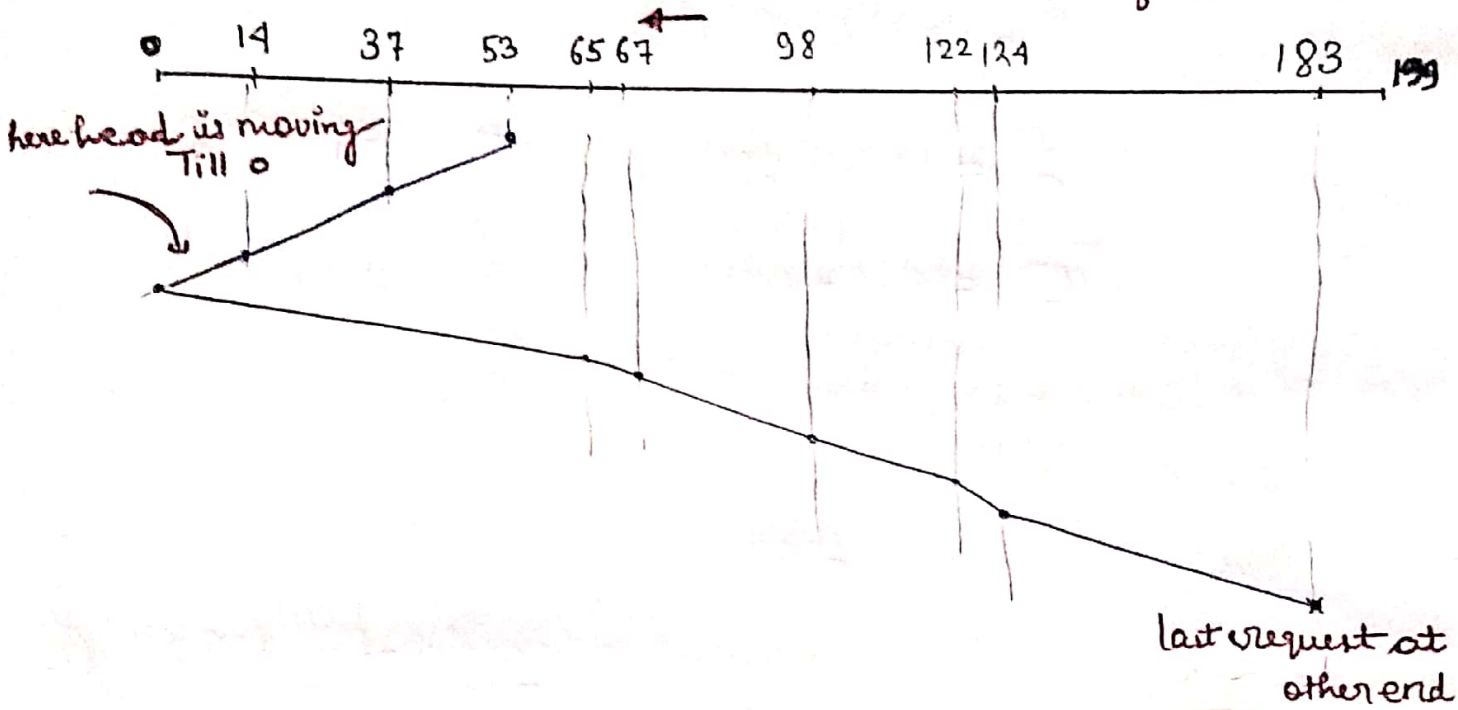
② Request arriving just behind will have to wait of reverse movement of Read-write head.

↑ <u>DRAWBACK</u>

Queue = 98, 183, 37, 122, 14, 124, 65, 67

Head starts at 53

• Direction of head is towards 0



④ C-SCAN Scheduling

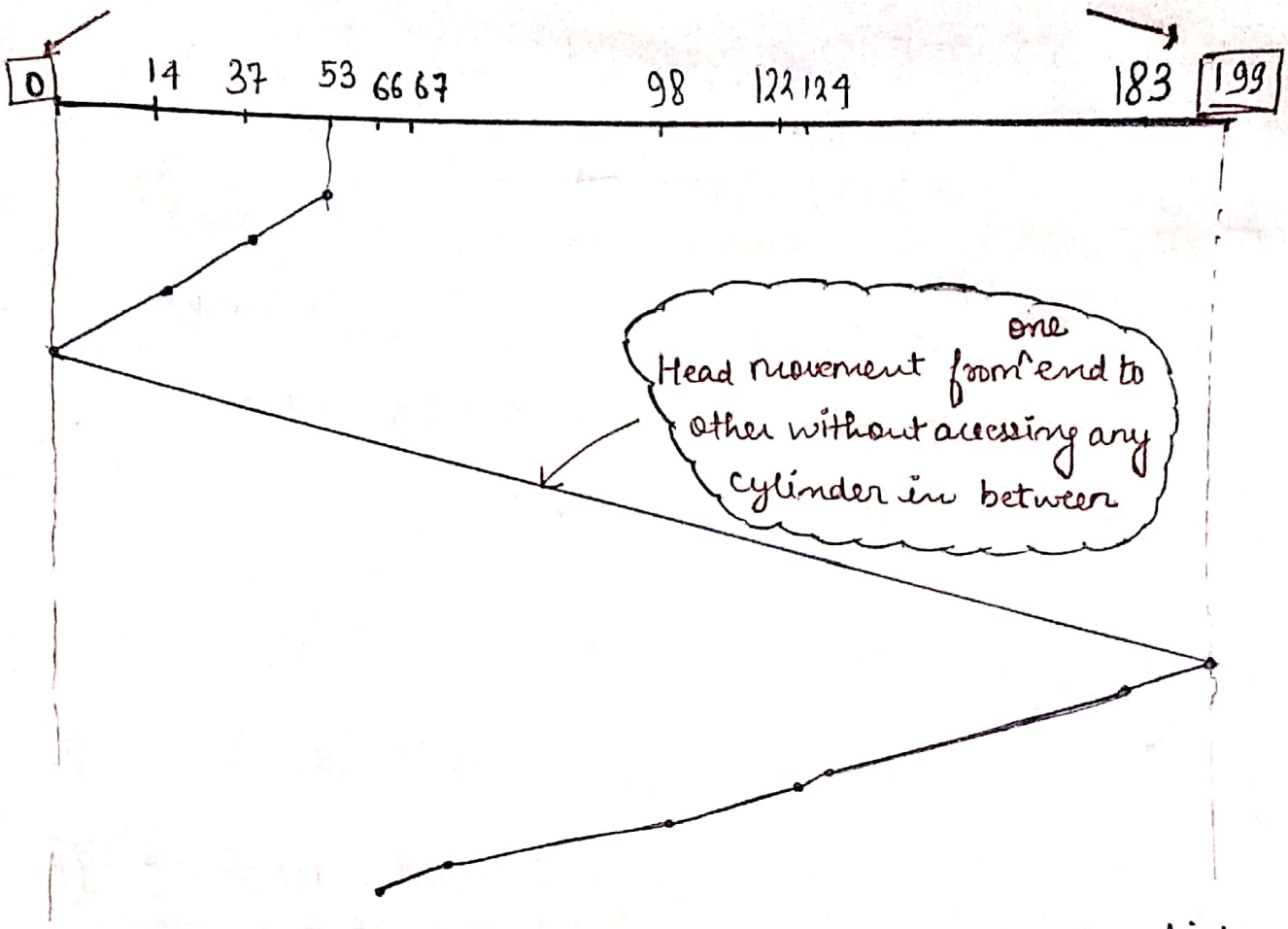
• Variant of SCAN scheduling

- Provide a uniform wait (No Request would be for longer time as in other case)
- When the read-write head reaches other end, it immediately return to the beginning or starting of disk without servicing any requests on the return trip.

Queue = 98, 183, 37, 122, 14, 124, 65, 67

Head starts at 53

• Direction of head is towards 0



In C-SCAN cylinders are treated as circular list.

(5) LOOK Scheduling

- In practice disk arm only moves to the final request in each direction.
- Then, direction of read-write head is reversed after servicing the final request.
- Versions of SCAN & C-SCAN that follows this procedure are called LOOK Scheduling.

Note: Here we are considering <u>total head movement</u>; But calculating Total <u>seektime</u> would require prior knowledge of time that is required in moving from one cylinder to other.

(i.e.) For FCFS in the given example

If time of movement = 2 nano seconds

Then, Total seektime = 640 × 2

= 1280 nano seconds

Seek time = Total Cylinder Access × Time to move between Adjacent Cylinders

Points to Ponder

- In SCAN Algorithm – Movement starts from one end of disk (say 0) till requested.
- C-SCAN Algorithm – Movement starts from one end of disk (say 0) to another end (say 199)
- LOOK Algorithm – Serves only requested cylinder either in SCAN manner or C-SCAN manner